

Employee Data Cleaning & Integrity Analysis

A Data Quality Case Study — Cleaning, Imputation & Critical Verification in Excel

Author: Franklin Manuel Ruiz Guadamuz · Tools: Advanced Excel (formulas, PivotTables, formula-based conditional formatting) · Source: HR dataset — 1,020 employee records

1. OBJECTIVE

Audit, clean, and analyze a raw employee dataset end to end — and, beyond standard cleaning, apply the techniques a real HR analytics task requires: imputing missing values responsibly, validating consistency across fields, and summarizing the result for decision-makers. The project's defining moment is not a dramatic discovery but its opposite: a striking finding that, on inspection, turned out to be an artifact. Catching that before reporting is the professional skill on display.

Employee ID	First Name	Last Name	Age	Department	Region	Status	Join Date	Salary	Email	Phone	Performance Score	Remote Work
EMP1000	Bob	Davis	25	DevOps	California	Active	4/20/2021	8807.65	bob.davis@example.com	-1652623197	Average	TRUE
EMP1001	Bob	Brown		Finance	Texas	Active	7/10/2020	6304.66	bob.brown@example.com	-1898471390	Excellent	TRUE
EMP1002	Alice	Jones		Admin	Nevada	Pending	12/7/2023	88145.9	alice.jones@example.com	-0506303711	Good	TRUE
EMP1003	Eva	Davis	35	Admin	Nevada	Inactive	11/27/2001	55458.99	eva.davis@example.com	-0476459780	Good	TRUE
EMP1004	Frank	Williams	25	Cloud Tech	Florida	Active	1/5/2022	109524.81	frank.williams@example.com	-1586734296	Poor	FALSE
EMP1005	Alice	Garcia	40	Sales	Texas	Inactive	6/10/2020	66642.84	alice.garcia@example.com	-5469033485	Good	FALSE
EMP1006	Frank	Jones		Admin	Nevada	Active	4/3/2020	84508.43	frank.jones@example.com	-4816379683	Good	FALSE
EMP1007	Bob	Jones	20	Cloud Tech	Florida	Inactive	7/17/2022	84457.91	bob.jones@example.com	-4114177589	Average	TRUE
EMP1008	Frank	Davis	38	Admin	Nevada	Inactive	12/9/2023	115963.82	frank.davis@example.com	-4177969133	Excellent	TRUE
EMP1009	Charlie	Johnson		DevOps	New York	Active	8/4/2022	78561.88	charlie.johnson@example.com	-6150950293	Excellent	TRUE
EMP1010	Daniel	Williams	35	Cloud Tech	New York	Active	7/6/2004	72885.77	daniel.williams@example.com	-3720634390	Average	FALSE
EMP1011	Daniel	Davis		DevOps	California	Active	9/4/2023	64844.19	daniel.davis@example.com	-1238294576	Poor	TRUE
EMP1012	Heidi	Williams	30	HR	New York	Active	3/5/2024	88266.77	heidi.williams@example.com	-3429643184	Excellent	FALSE
EMP1013	Grace	Williams		Finance	California	Active	10/10/2020	65666.84	grace.williams@example.com	-0463038882	Good	FALSE
EMP1014	Alice	Garcia	25	Admin	Nevada	Active	6/10/2024	97633.68	alice.garcia@example.com	-1734236360	Excellent	TRUE
EMP1015	Grace	Jones	40	Sales	Florida	Pending	5/25/2021	117575.43	grace.jones@example.com	-4015251385	Good	TRUE
EMP1016	Alice	Brown		DevOps	Illinois	Inactive	4/23/2024	54867.39	alice.brown@example.com	-7713334475	Poor	FALSE
EMP1017	Grace	Garcia	30	Finance	Texas	Pending	11/26/2024	100377.85	grace.garcia@example.com	-2639043825	Average	TRUE
EMP1018	Charlie	Garcia		Cloud Tech	Nevada	Inactive	6/19/2024	84726.46	charlie.garcia@example.com	-0844387398	Excellent	FALSE
EMP1019	Charlie	Williams	35	Admin	Illinois	Pending	1/20/2020	91826.12	charlie.williams@example.com	-7107236265	Average	TRUE
EMP1020	Bob	Miller	35	HR	New York	Active	5/6/2021	111462.67	bob.miller@example.com	-6431808336	Excellent	TRUE
EMP1021	Bob	Jones	40	Cloud Tech	Illinois	Active	5/15/2020	50382.56	bob.jones@example.com	-0590048901	Good	FALSE
EMP1022	Charlie	Johnson	35	Sales	Illinois	Inactive	3/9/2021	115823.94	charlie.johnson@example.com	-6528924227	Average	TRUE
EMP1023	Daniel	Garcia		Finance	Nevada	Pending	2/3/2022	60667.01	daniel.garcia@example.com	-7572158623	Excellent	FALSE
EMP1024	Bob	Brown	30	Admin	Texas	Inactive	2/13/2020	91181.61	bob.brown@example.com	-1562749543	Good	TRUE
EMP1025	Daniel	Miller	25	Cloud Tech	Nevada	Pending	4/30/2020	64628.82	daniel.miller@example.com	-6480425727	Average	FALSE
EMP1026	Frank	Miller	25	Sales	New York	Inactive	10/9/2022	63079.01	frank.miller@example.com	-7003032073	Average	FALSE
EMP1027	Heidi	Johnson	30	Finance	Florida	Active	7/10/2020	56785.16	heidi.johnson@example.com	-2761707580	Excellent	TRUE
EMP1028	Grace	Garcia	40	Finance	Nevada	Active	10/11/2023	95741.86	grace.garcia@example.com	-7712515341	Excellent	FALSE
EMP1029	Alice	Brown	40	DevOps	California	Active	5/24/2023	63413.31	alice.brown@example.com	-2849838988	Excellent	TRUE
EMP1030	Charlie	Jones	48	Admin	California	Inactive	5/26/2023	111224.02	charlie.jones@example.com	-2613719340	Good	TRUE
EMP1031	Bob	Smith	25	DevOps	Illinois	Active	7/10/2020	114843.69	bob.smith@example.com	-6361528176	Excellent	TRUE
EMP1032	Charlie	Brown	40	Finance	Florida	Active	11/5/2024	78072.82	charlie.brown@example.com	-7420639739	Average	FALSE

Before — raw export: a combined Department-Region field, US-format dates, sign-prefixed phone numbers, missing ages, and corrupt salary values (top rows shown)

2. THE CENTRAL FINDING — A RESULT WORTH DOUBTING

A cross-field integrity check compared each employee's tenure (calculated from their join date) against their age, on the simple premise that no one starts working before about sixteen. The first pass was alarming: **190 records — roughly one in five — appeared logically impossible.**

Rather than report that number, it was pressure-tested. The cause was mechanical: the 211 missing ages had been temporarily filled with zero, and $\text{tenure} > 0 - 16$ is true for everyone, so every age-less row looked impossible. After imputing the missing ages properly and re-running the check against the **original recorded ages**, the count of real impossibilities fell to **zero** — employees with a recorded age are internally consistent (minimum recorded age 25, maximum tenure 6 years).

The real integrity issue is quieter. 21% of records have no age at all, and the ages that do exist appear only in 5-year increments (25 / 30 / 35 / 40). Together these indicate that Age was generated synthetically and independently of the join date — a genuine limitation for any age-based analysis, and a more honest headline than "190 impossible employees."

3. RESPONSIBLE IMPUTATION

The 211 missing ages were not deleted, nor blandly labeled "missing." Each was imputed with the **average age of the employee's own department** (AVERAGEIF, rounded to a whole number) — a more defensible estimate than a single global average. Critically, an **Age_Was_Imputed** flag marks every estimated value, so downstream users always know which ages are real and which are inferred. The integrity check above was deliberately run against the *original* ages, not the imputed ones, so that a genuine inconsistency could never be masked by an estimate.

Age	Class	Age_Year_Imputed	Department	Class	Region	Class	Status	Class	Join_Date	Class	Salary	Class	Phone	Class	Performance	Class	Remote_Work	Class	Tenure	Year	Age_Tenure	Class	Age_raw_check
75	No		DevOps		California		Active		2021-08-02		181767.65		1511521197		Average		True		3	OK		OK	
32	Yes		Finance		Texas		Active		2020-07-10		653045.66		18156471390		Excellent		True		0	No_Age		No_Age	
52	Yes		Admin		Nevada		Pending		2023-12-07		88145.9		5595385211		Good		True		2	No_Age		No_Age	
26	No		Admin		Nevada		Inactive		2023-11-27		69450.99		5478497784		Good		True		4	OK		OK	
26	No		Cloud Tech		Florida		Active		2022-01-05		1093204.61		1586734256		Poor		False		4	OK		OK	
40	No		Sales		Texas		Inactive		2020-06-10		80642.84		545993485		Good		False		6	OK		OK	
27	Yes		Admin		Nevada		Active		2020-04-03		90700.43		4318175063		Good		False		6	No_Age		No_Age	
30	No		Cloud Tech		Florida		Inactive		2022-07-17		36397.36		4134327309		Average		True		4	OK		OK	
35	No		Admin		Nevada		Inactive		2023-12-09		1093204.61		1178561123		Excellent		True		2	OK		OK	
33	Yes		DevOps		New York		Active		2022-09-26		76561.88		6556895899		Excellent		True		3	No_Age		No_Age	
36	No		Cloud Tech		New York		Active		2024-07-06		72061.71		2729634590		Average		False		2	OK		OK	
33	Yes		DevOps		California		Active		2023-09-04		83846.19		1238296376		Poor		True		2	No_Age		No_Age	
30	No		HR		New York		Active		2024-03-03		80795.77		2429543164		Excellent		False		2	OK		OK	
32	Yes		Finance		California		Active		2020-10-10		48428.84		2603303802		Good		False		5	No_Age		No_Age	
25	No		Admin		Nevada		Active		2024-06-10		97053.88		1734209968		Excellent		True		2	OK		OK	
40	No		Sales		Florida		Pending		2022-05-29		112975.48		4215291065		Good		True		5	OK		OK	
26	No		DevOps		Illinois		Inactive		2024-04-23		94967.33		7713334475		Poor		False		2	OK		OK	
30	No		Finance		Texas		Pending		2024-11-26		100377.85		2636342625		Average		True		1	OK		OK	
27	No		Cloud Tech		Nevada		Inactive		2024-09-19		94776.45		2644230739		Terrible		False		1	No_Age		No_Age	
26	No		Admin		Illinois		Pending		2020-01-02		91109.12		7107293305		Average		True		6	OK		OK	
35	No		HR		New York		Active		2023-05-08		111860.47		9411803536		Excellent		True		5	OK		OK	
40	No		Cloud Tech		Illinois		Active		2020-09-19		96182.58		5898049963		Good		False		5	OK		OK	
36	No		Sales		Illinois		Inactive		2021-03-09		115025.84		682892427		Average		True		5	OK		OK	
36	No		Finance		Nevada		Pending		2022-02-03		80667.81		7872159623		Excellent		False		4	OK		OK	
30	No		Admin		Texas		Inactive		2020-07-13		91161.61		1102745643		Good		True		6	OK		OK	
25	No		Cloud Tech		Nevada		Pending		2020-06-30		60025.32		5485021727		Average		False		6	OK		OK	
25	No		Sales		New York		Inactive		2023-10-02		83979.81		7652532973		Average		False		2	OK		OK	
30	No		Finance		Florida		Active		2020-07-12		96785.48		2784297596		Excellent		True		6	OK		OK	
30	No		Finance		Nevada		Active		2020-10-11		96741.88		7712151341		Excellent		False		2	OK		OK	
40	No		DevOps		California		Active		2021-05-34		63431.31		2846839908		Excellent		True		5	OK		OK	
40	No		Admin		California		Inactive		2021-05-26		111214.82		2613715949		Good		True		3	OK		OK	
25	No		DevOps		Illinois		Active		2020-07-10		116945.65		65051520176		Excellent		True		6	OK		OK	
40	No		Finance		Florida		Active		2024-11-05		78075.82		7158509758		Average		False		1	OK		OK	
25	No		Finance		California		Inactive		2021-08-07		79810.82		2354501895		Average		False		4	OK		OK	
30	No		Admin		California		Active		2020-04-29		90542.42		2842381931		Excellent		True		6	OK		OK	
26	No		Cloud Tech		Illinois		Inactive		2022-10-26		114307.1		6642147649		Excellent		True		3	OK		OK	

Cleaning layer — raw and cleaned columns side by side; salaries more than 35% above their own department's average are highlighted by a formula-based rule (top rows shown)

4. CLEANING SUMMARY

Issue	Column	Resolution
Two attributes in one field	Department_Region	Split into Department and Region (TEXTSPLIT / TEXTAFTER)
Regional date misparse risk	Join_Date	Parsed explicitly and standardized to ISO YYYY-MM-DD
Spurious leading sign	Phone	Removed the stray minus sign; digits left untouched
Corrupt values	Salary	24 "Invalid" entries flagged and blanked to keep the column numeric
Decimal notation	Salary	US period converted to regional separator before numeric use
Category variants	Status / Performance / Remote_Work	Verified clean against known value sets (TRIM + PROPER + UNIQUE)

Check	Comment	Results affected	Severity	Fixing a Denial
1. Duplicates OK	Employees_ID	0	Low	Verified uniqueness with COUNTA in COUNTA(UNIQUE). All 1,000 OK.
2. Blank values	Employees_ID	0	Low	No blanks found (COUNTIF for empty strings).
3. Blank values	Age	211	High	211 blank Age values (~21%) Imputed using the department average rather than deleted, and flagged with an Age_Year_Imputed column for full traceability.
4. Blank values	First_Name / Last_Name / Department / Region / Status / Join_Date / Salary / Email / Phone / Performance	0	Low	No blanks found in the remaining columns (COUNTIF per column).
5. Leading/trailing spaces	First_Name / Last_Name	0	Low	Confirmed 0 blanks in either TRIM — no differences, no stray spaces.
6. Leading/trailing spaces	Phone / Remote_Work	0	Low	Re-checked after correcting an initial false positive (a date type artifact, not real spaces).
7. Null / numeric values	All columns	0	Low	Scanned for full valid (COUNTIF for null/not null — none blank).
8. Non-numeric values	Salary	24	High	24 salary values contained the text "Invalid" instead of a number. Flagged and left blank in the deliverable so the column stays numeric and can be aggregated.
9. Decimal notation (period separator)	Salary	Summing	Medium	US decimal notation converted to the regional separator before numeric use (see below for format).
10. Inconsistent date format	Join_Date	Transformation	Medium	All dates in US/MD/YYYY format. Parsed explicitly (LEFT(MID(RIGHT + DATE) and standardized to ISO YYYY-MM-DD to avoid regional misparsing.
11. Combined field into attributes (more columns)	Department_Region	Transformation	Low	Column split into Department and Region (e.g., DevOps California). Set into Department and Region with TEXTSPLIT / TEXTAFTER.
12. Data type validation	Age	0	Low	Confirmed values are numeric (ISNUMBER across the column).
13. Phone number formatting	Phone	1020	Medium	All phone values carried a leading minus sign. Removed (TRIM + LEFT + RIGHT) to leave a clean number. Could not confirm whether a country code is present, so only the stray symbol was removed.
14. Value range check (synthetic data signal)	Age	0	Low	Age contains only 0-year increments (20/20/20/0) — a sign of synthetic or heavily rounded data. Limits granular age analysis. Documented, not altered.
15. Future / impossible dates	Join_Date	0	Medium	Checked for join dates in the future (IF > TODAY). None found — all join dates are plausible.
16. Category consistency	Status	0	Low	TRIM + PROPER, then UNIQUE. Three clean values: Active, Pending, Inactive.
17. Category consistency	Performance_Score	0	Low	TRIM + PROPER, then UNIQUE. Four clean values: Good, Excellent, Average, Poor.
18. Category consistency	Remote_Work	0	Low	TRIM + PROPER, then UNIQUE. Three clean values: True, False.
19. Value range check	Salary	0	Medium	Average valid values: Min 30,027 / Max 119,372 / Median 85,546. No negative or zero salaries.
20. Cross-field integrity (Age vs tenure)	Age / Join_Date	0 (raw)	Critical	Calculated tenure (YEARFrac of join date to today) and compared against Age. Identified each Age started with no more than 10. An initial pass flagged 180 records as "Impossible", but investigation showed this was an ARTIFACT of zero flagging the 211 missing ages (tenure > 10 is always true), NOT a genuine inconsistency. Re-running the check against the ORIGINAL, unrounded Age produced zero real impossible/illegal employees with a rounded age an internally consistent (zero rounded age 10, max tenure 6 years). The genuine integrity issue in the 211 records (21% back log entries, and ages appear only 10-year increments — indicating Age was generated synthetically and independently of Join_Date. Key lesson: a string flag was more useful before reporting rather than taken at face value.

Audit log — 20 checks catalogued with severity and the decision taken for each

5. ANALYSIS LAYER

Three PivotTables turn the cleaned data into a decision-ready summary, and a formula-based conditional format surfaces salaries that are high *relative to their own department* — a within-group comparison that Excel's built-in rules cannot express.

Headcount & Avg Salary by Department and Status									
Column Labels									
Row Labels	Active		Inactive		Pending		Total Employees		Total Average Salary
	Employees	Average Salary	Employees	Average Salary	Employees	Average Salary			
Admin	58	84347,37696	51	85343,29563	57	85894,18018	166		85191,9228
Cloud Tech	45	84285,02814	42	87679,79439	59	83421,4093	146		84923,03624
DevOps	78	86499,98141	58	86039,44604	53	85280,36151	189		86016,02364
Finance	49	83863,67979	62	80617,95097	59	82652,82441	170		82240,61042
HR	57	84730,54127	49	83471,37489	65	87550,21569	171		85473,64317
Sales	65	85982,14406	50	85258,46816	63	89135,43581	178		86896,68103
Grand Total	352	85129,26475	312	84501,90973	356	85735,19955	1020		85155,0564
Performance Distribution by Region						Data Integrity: Age Availability by Department			
Region	Column Labels						Department	Column Labels	
Row Labels	Average		Excellent	Good	Poor	Grand Total	Row Labels	Ok	No Age
									Grand Total
California	48		51		51	37	Admin		129
Florida	50		37		47	51	Cloud Tech		110
Illinois	45		45		43	32	DevOps		146
Nevada	43		52		46	28	Finance		142
New York	38		47		38	38	HR		137
Texas	43		35		45	30	Sales		145
Grand Total	267		267		270	216	Grand Total		809
						1020			211
									1020

Analysis — headcount and average salary by department and status, performance distribution by region, and age availability by department

6. GUIDING PRINCIPLES

- **Doubt your most striking result.** The 190 "impossibilities" were verified before being reported — and proved to be an artifact.
- **Impute transparently.** Estimated values are flagged, never silently mixed with real ones.
- **Validate integrity on real data, not estimates.** The check ran against original ages so imputation couldn't hide a problem.
- **Distinguish exploration from error-detection.** The salary highlight is an analytical lens, not a data-quality alarm — and is described as such.

7. RESULTS

Metric	Value
Records	1,020
Checks catalogued	20 (1 critical, 2 high, 6 medium, 11 low)
Missing ages imputed and flagged	211
Corrupt salaries flagged	24
Real age-tenure impossibilities	0 (the initial 190 were an artifact)
New techniques applied	Group imputation, tenure calculation, PivotTables, formula-based conditional formatting

Deliverables: a cleaned dataset with an imputation flag and an integrity flag per record, a 20-row audit log with severity and decisions, a fully auditable formula layer, and a PivotTable analysis summary.